

Identifying & Analyzing Malicious Activity (CS0-003)

Introduction

Objective

Welcome to the **Identifying & Analyzing Malicious Activity** practice lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Overview

Learning Outcomes:

In this module, you will complete the following exercises:

- Exercise 1 – Analyze Resource Utilization
- Exercise 2 – Detecting Unauthorized User Privilege Escalation

After completing this module, you should be able to:

- Monitor Resource Utilization on a Windows Device
- Monitor Resource Utilization on a Linux Device
- Enable User Privilege Monitoring
- Generate Auditing Logs
- Detect an Unauthorized User Privilege Escalation

Exam Objectives:

The following exam objectives are covered in this module:

1.2 Given a scenario, analyze indicators of potentially malicious activity.

- Network-related
- Host-related

- Application-related

Exercise 1 – Analyze Resource Utilization

As a Cyber Security Specialist, it is essential to monitor the resource utilization of devices. A sudden increase in resource utilization, such as increased network traffic or processor consumption, may indicate a network-related attack. Windows Operating system has a native application that can be used to monitor resource utilization. Other tools are available that can be used to get more details about resource utilization.

In this exercise, different methods for monitoring network resources will be investigated.

Learning Outcomes

After completing this exercise, you should be able to:

- Monitor Resource Utilization on a Windows Device
- Monitor Resource Utilization on a Linux Device

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01** – Windows Server 2022 – Domain Controller
- **ACIDM01** – Windows Server 2022 – Domain Member Server
- **ACIALMA** – ALMA Linux Standalone Workstation

Task 1 – Monitor Resource Utilization on a Windows Device

A Cyber Security Specialist needs to monitor the resource utilization of devices on the network to determine and prevent malicious activity.

In this task, resource utilization will be monitored to detect malicious activity.



Click the **Start button**.



2 **Connect** to **ACIDM01**.

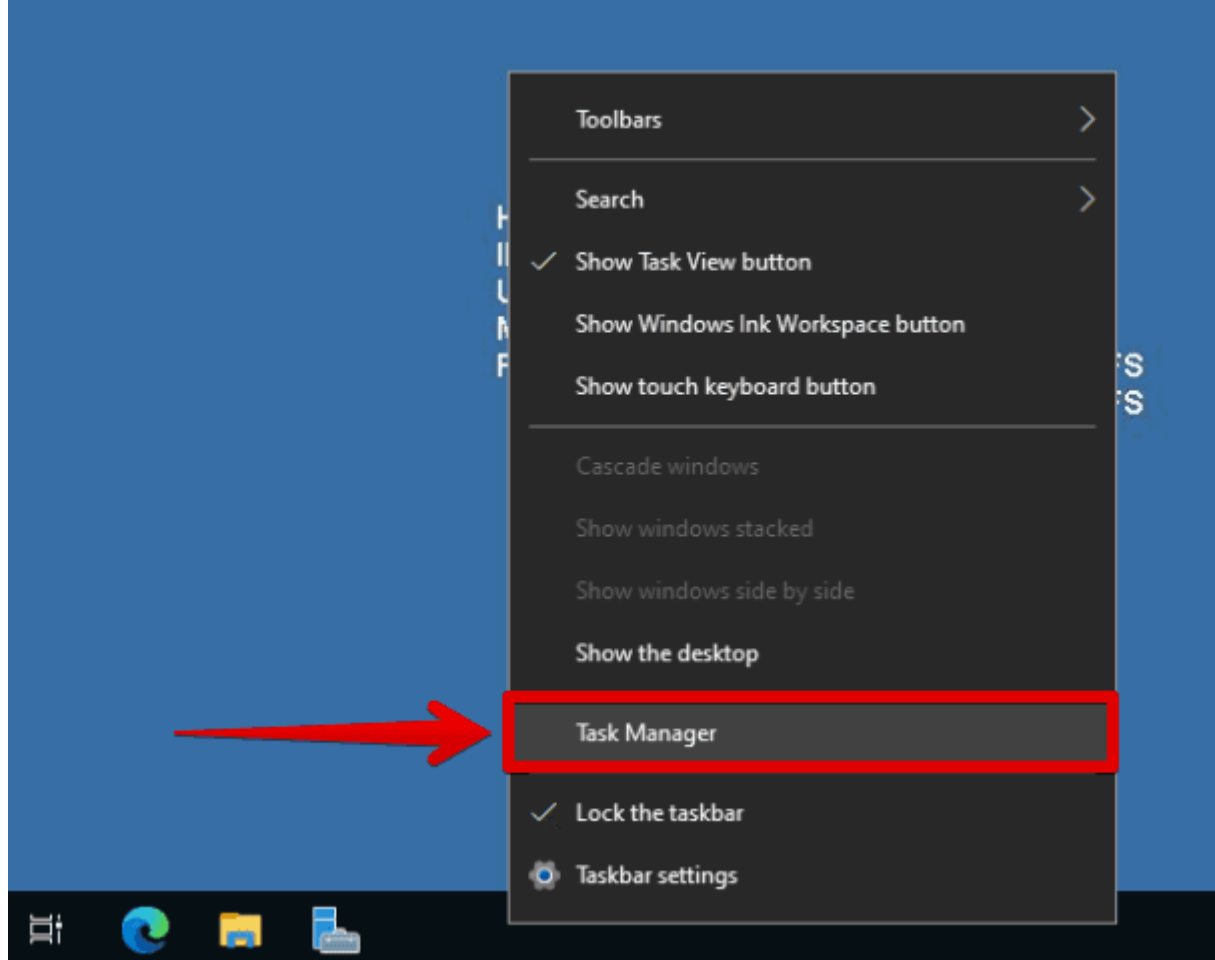
Click **Send Ctrl+Alt+Delete button** on the top of the screen.

Sign-in with the **following credentials**:

Username: **ACIPLAB\Administrator**

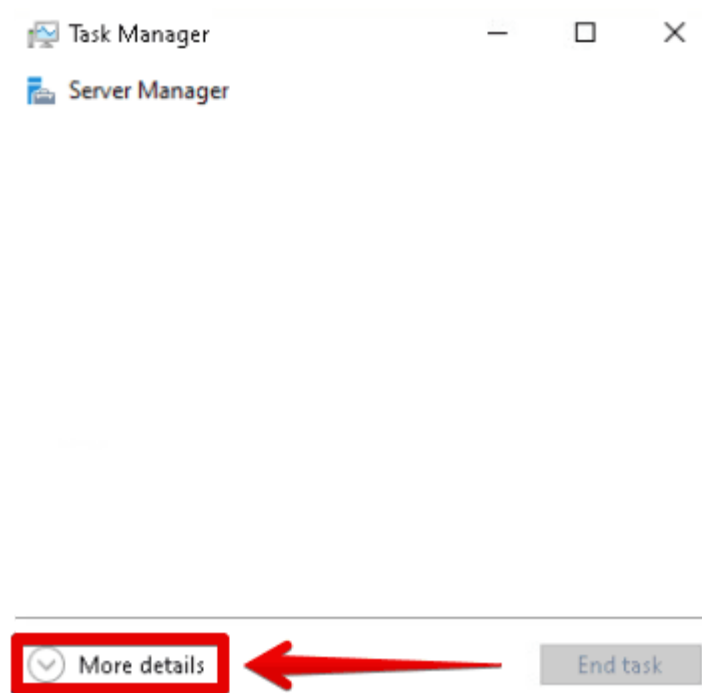
Password: **Passw0rd**

Right-click on the **Taskbar** and **select Task Manager**.



3

In the **Task Manager** window, **click** **More details**.



4

In the **Task Manager** window, **select** the **Performance** tab.

Task Manager

File Options View

Processes Performance Users Details Services

Name Status CPU Memory

Apps (2)

- > Server Manager 0% 46.1 MB
- > Task Manager 3.3% 19.1 MB

Background processes (33)

- AggregatorHost 0% 0.6 MB
- > Antimalware Service Executable 0% 99.7 MB
- COM Surrogate 0% 1.2 MB
- COM Surrogate 0% 1.3 MB
- > COM Surrogate 0% 3.1 MB
- CTF Loader 0% 2.9 MB
- Host Process for Windows Tasks 0% 1.9 MB
- > Hyper-V Host Compute Service 0% 1.3 MB
- > Microsoft Distributed Transaction... 0% 2.0 MB
- > Microsoft Network Realtime Ins... 0% 2.3 MB

^ Fewer details End task

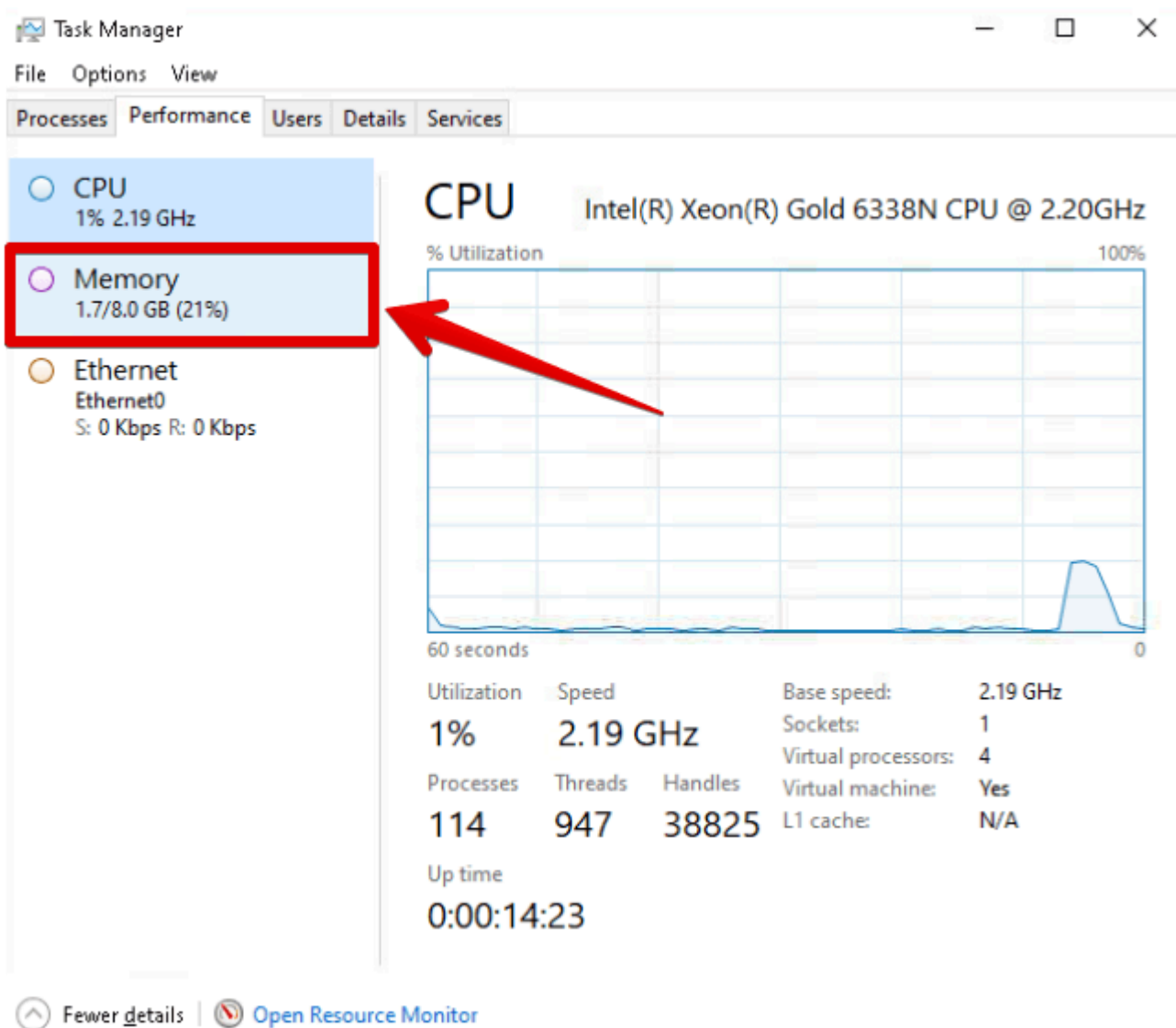
Name	Status	CPU	Memory
Apps (2)			
Server Manager		0%	46.1 MB
Task Manager		3.3%	19.1 MB
Background processes (33)			
AggregatorHost		0%	0.6 MB
Antimalware Service Executable		0%	99.7 MB
COM Surrogate		0%	1.2 MB
COM Surrogate		0%	1.3 MB
COM Surrogate		0%	3.1 MB
CTF Loader		0%	2.9 MB
Host Process for Windows Tasks		0%	1.9 MB
Hyper-V Host Compute Service		0%	1.3 MB
Microsoft Distributed Transaction...		0%	2.0 MB
Microsoft Network Realtime Ins...		0%	2.3 MB



Note: The CPU and Memory utilization can be seen in the Task Manager window. A sudden increase in resource utilization might indicate an attack on the device.

5

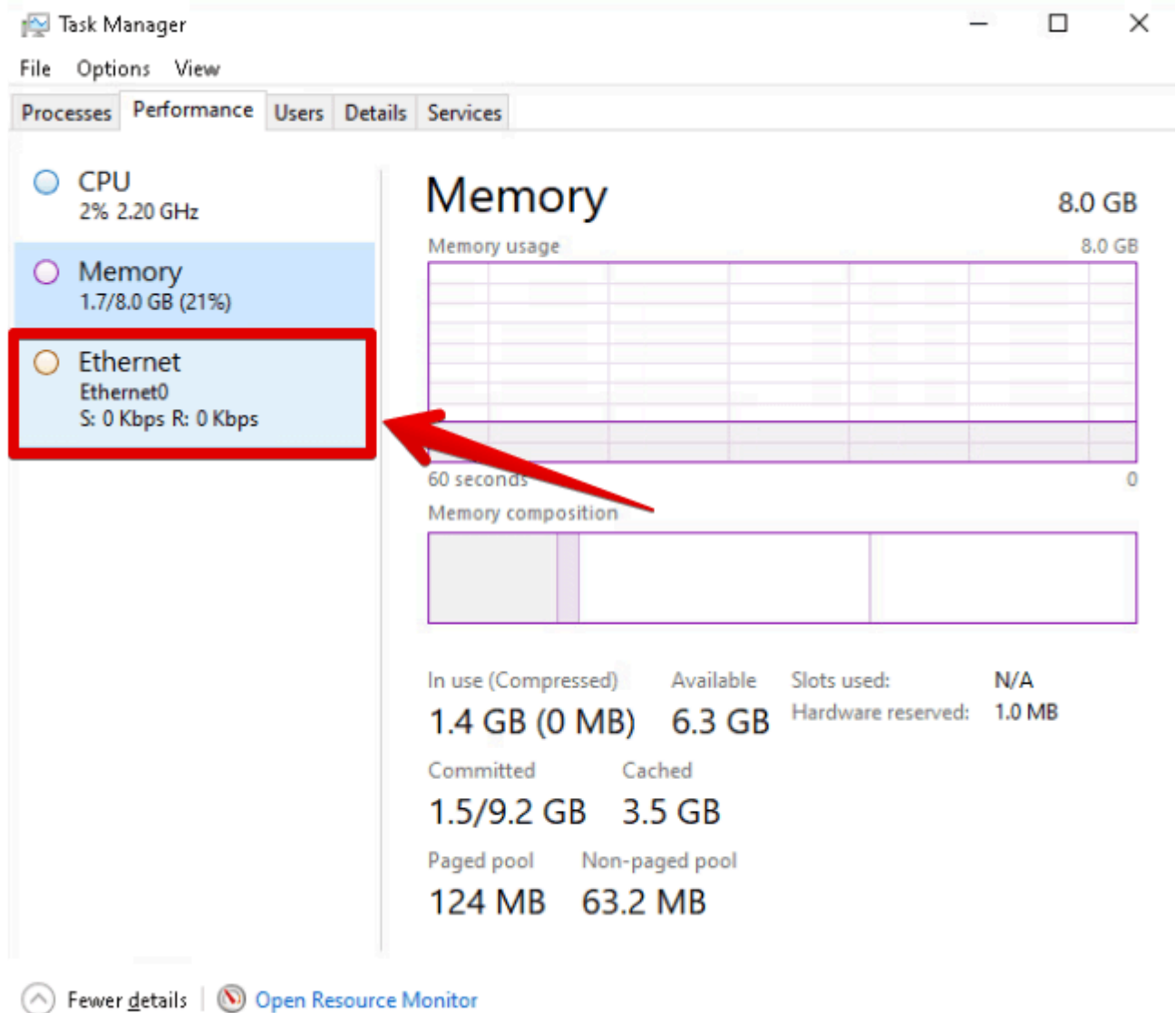
In the **Performance** tab, **select** **Memory**.



Note: A sudden increase in memory and CPU utilization might indicate that malware software is running on the device.

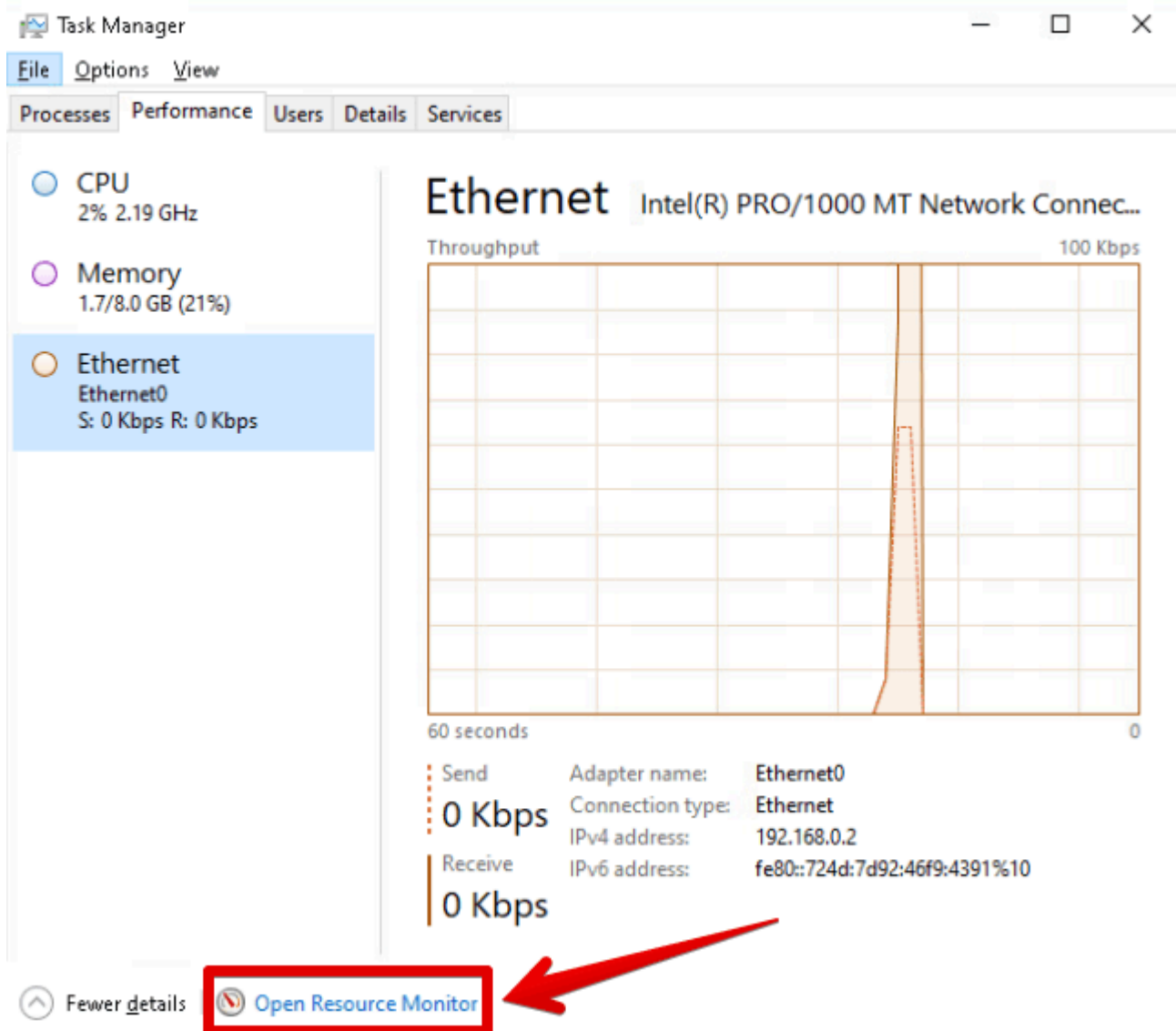
6

Click the **Ethernet** in the **Performance** tab.



7

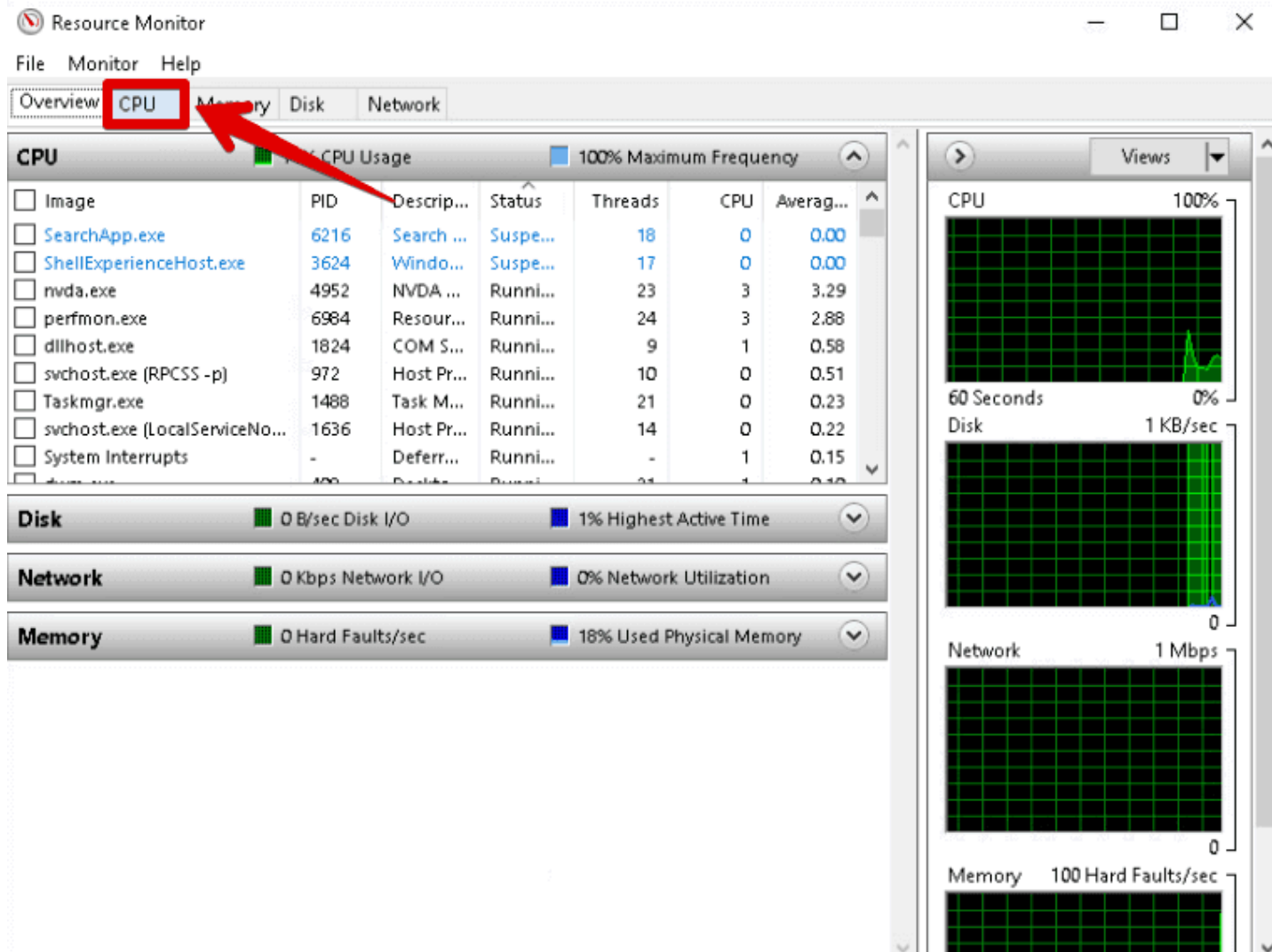
Click [Open Resource Monitor](#) in the [Task Manager](#) window.



Note: An increase in Ethernet utilization might indicate a Denial of Service attack on the device.

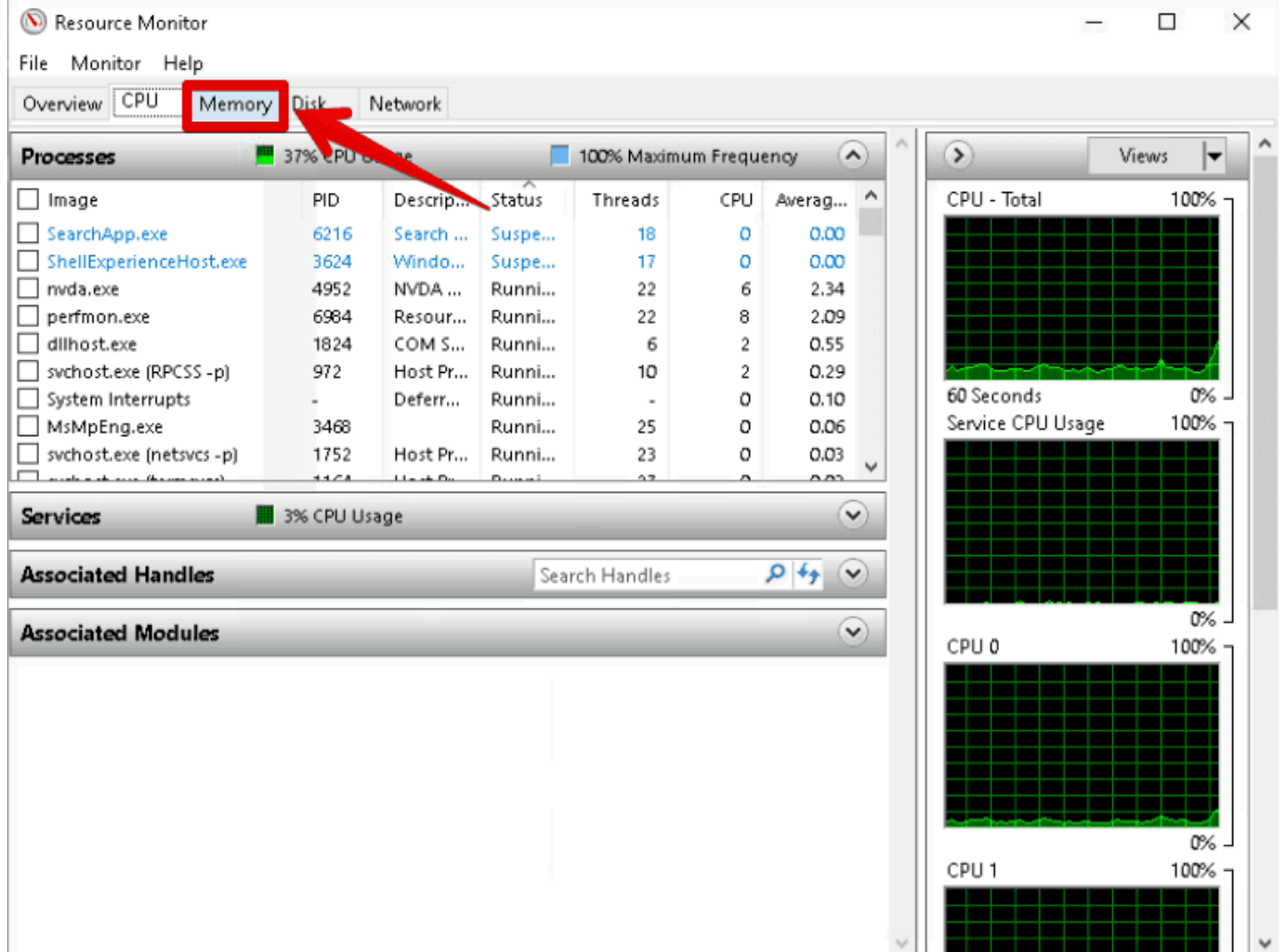
8

In **Resource Monitor**, **click** the **CPU** tab.



9

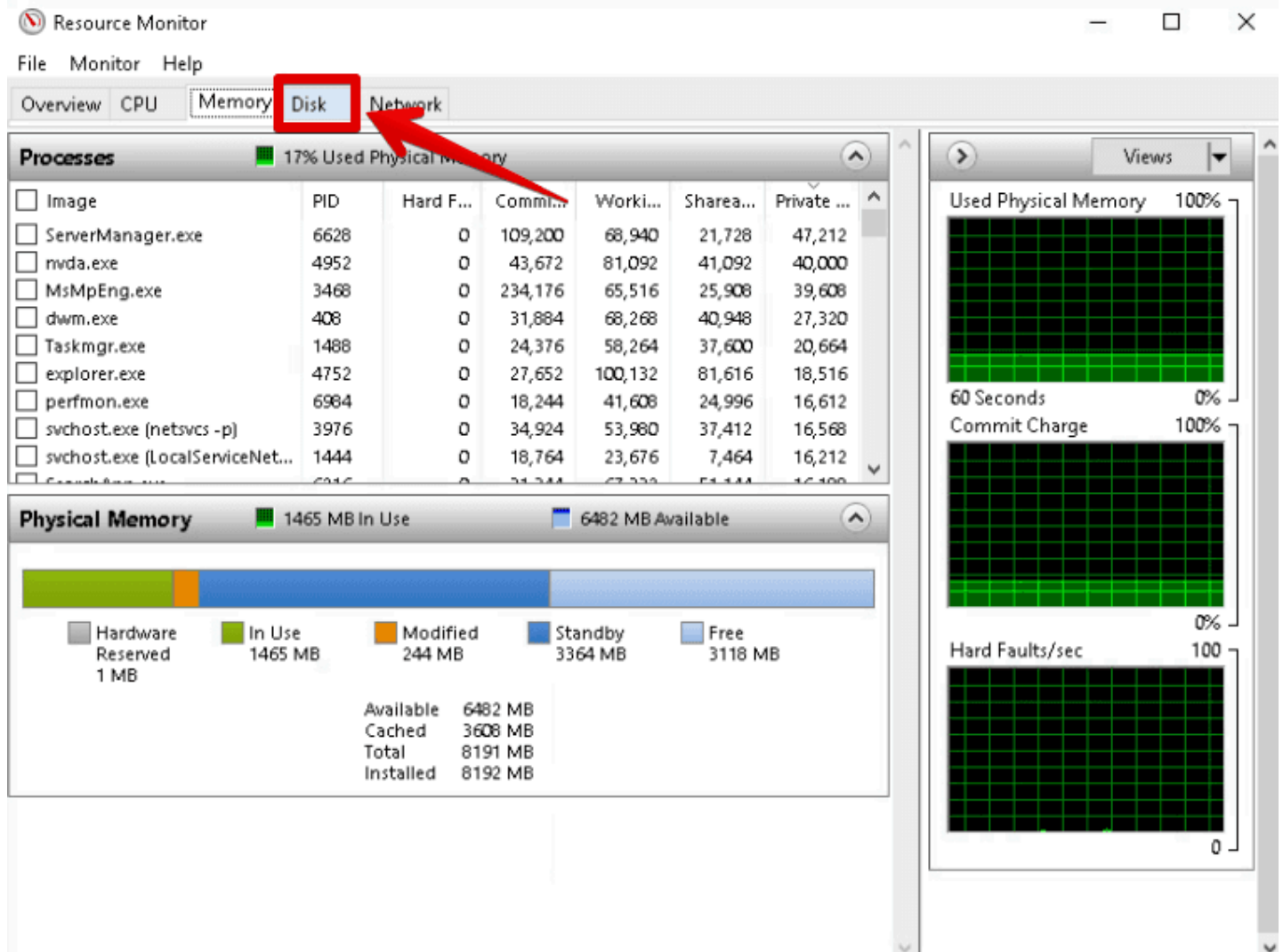
Click the Memory tab in Resource Monitor.



Note: More information about the different services running the device is displayed on the CPU tab. If a sudden increase in CPU utilization is noticed, malicious services can be detected here.

10

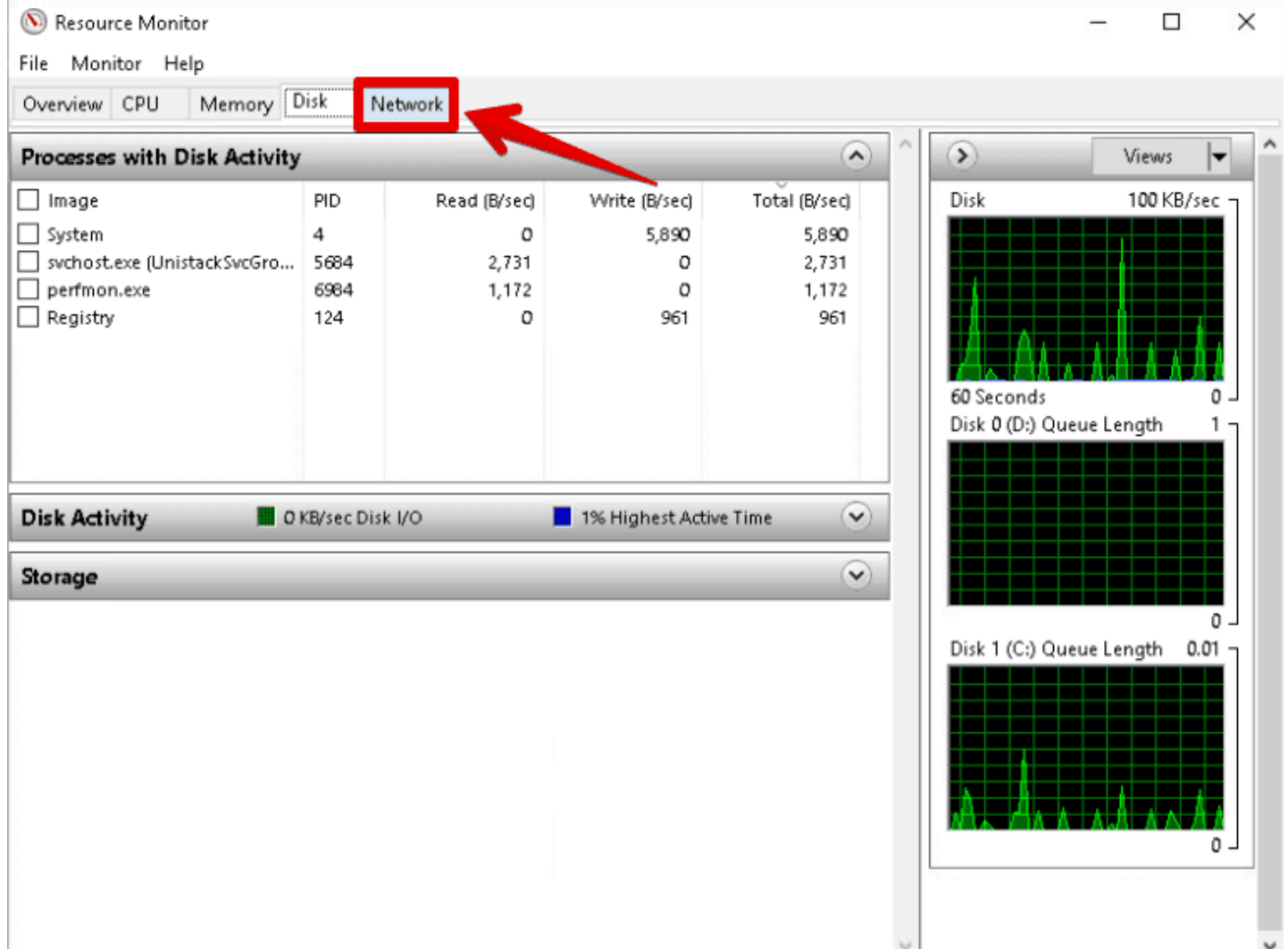
Select the **Disk** tab in the **Resource Monitor** window.



Note: The different services that utilize memory are displayed on the tab. Malicious software that consumes memory can be detected here and stopped.

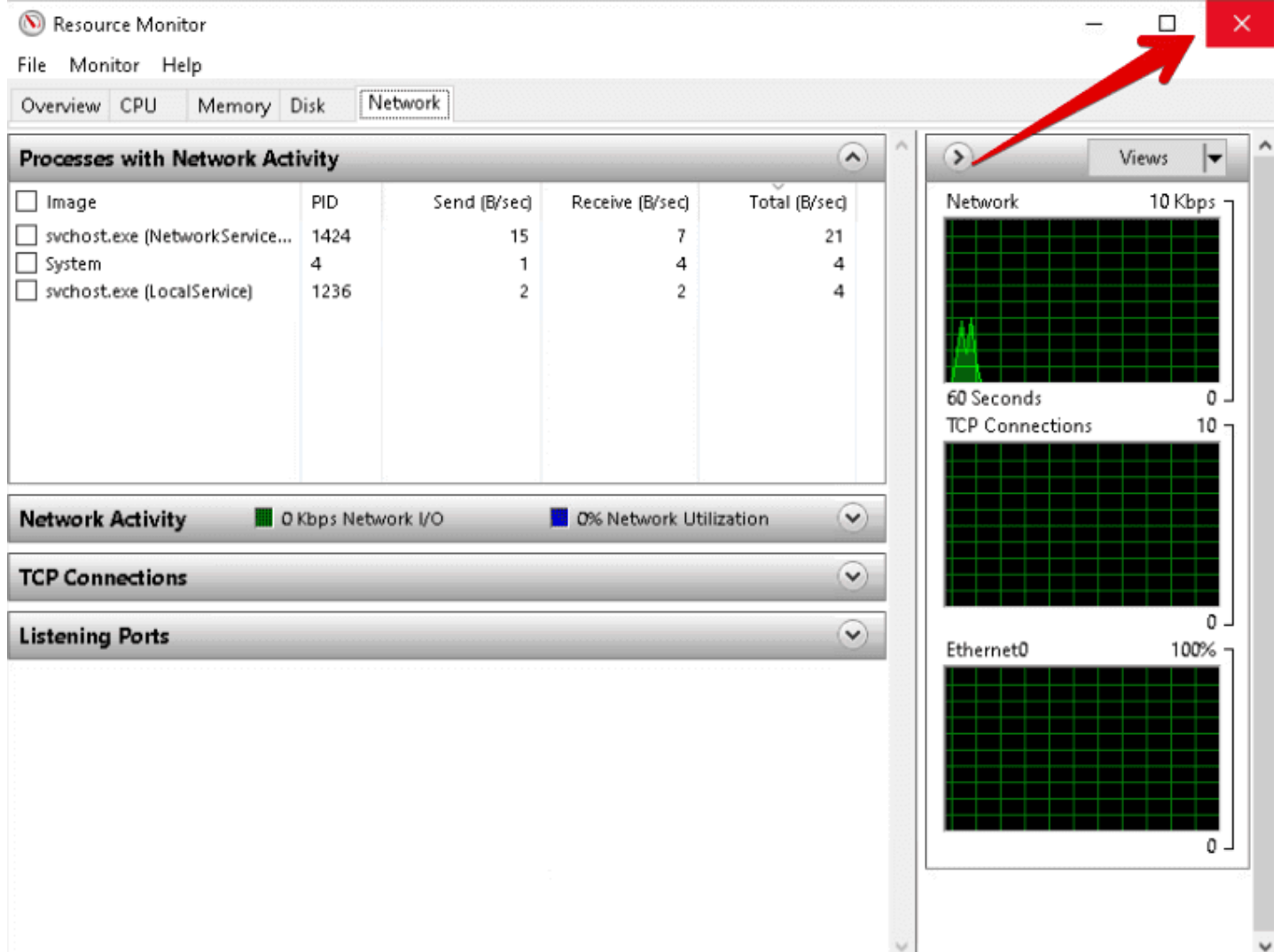
11

Select the **Network tab** in the **Resource Monitor** window.



12

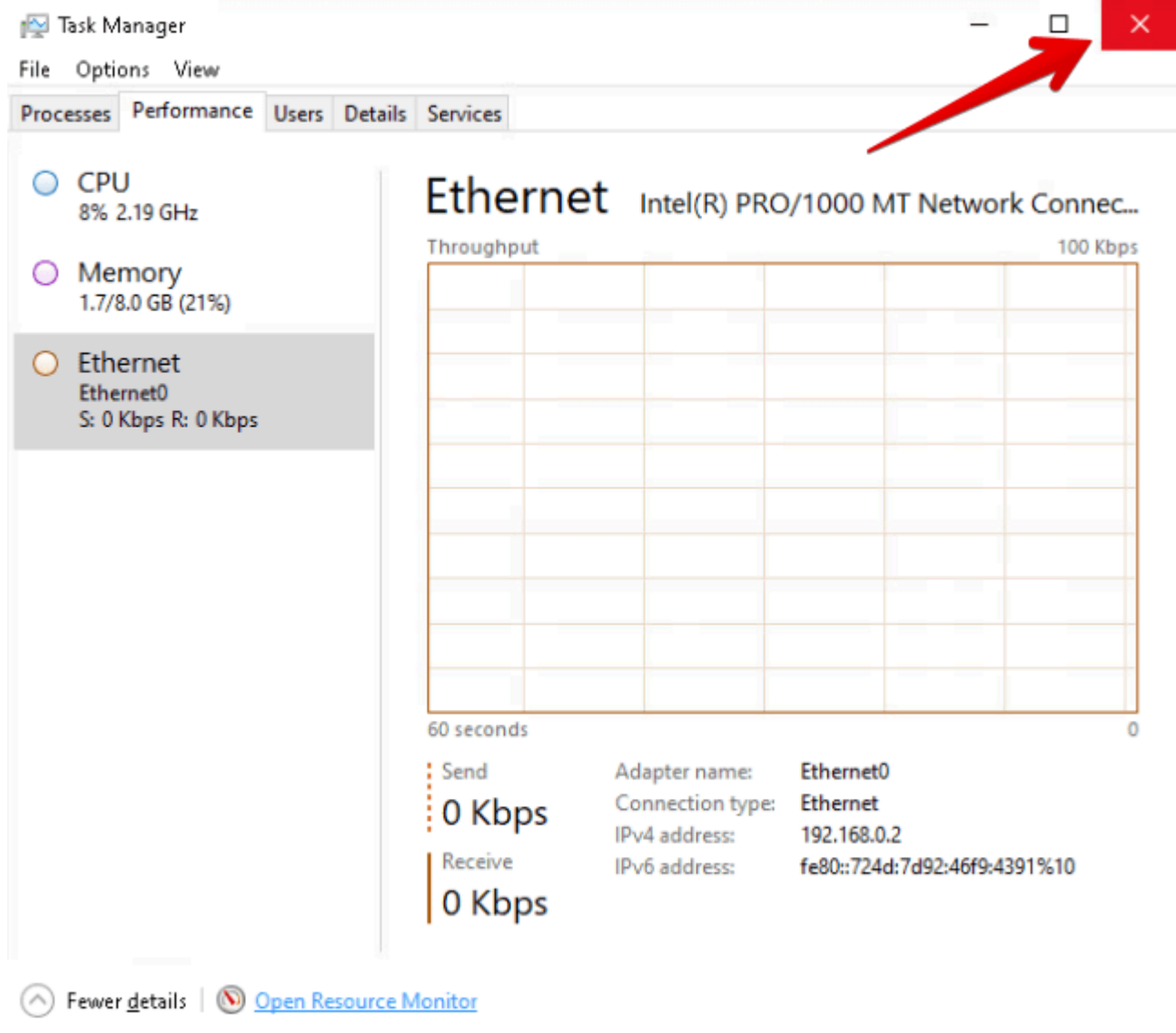
Close the Resource Monitor window.



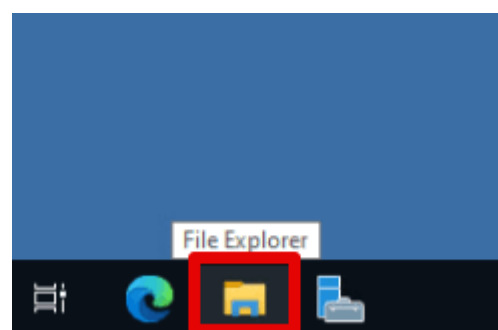
Note: On the **Network tab**, the different networking services that are running on the device can be seen. Malicious activity and services can be detected here.

13

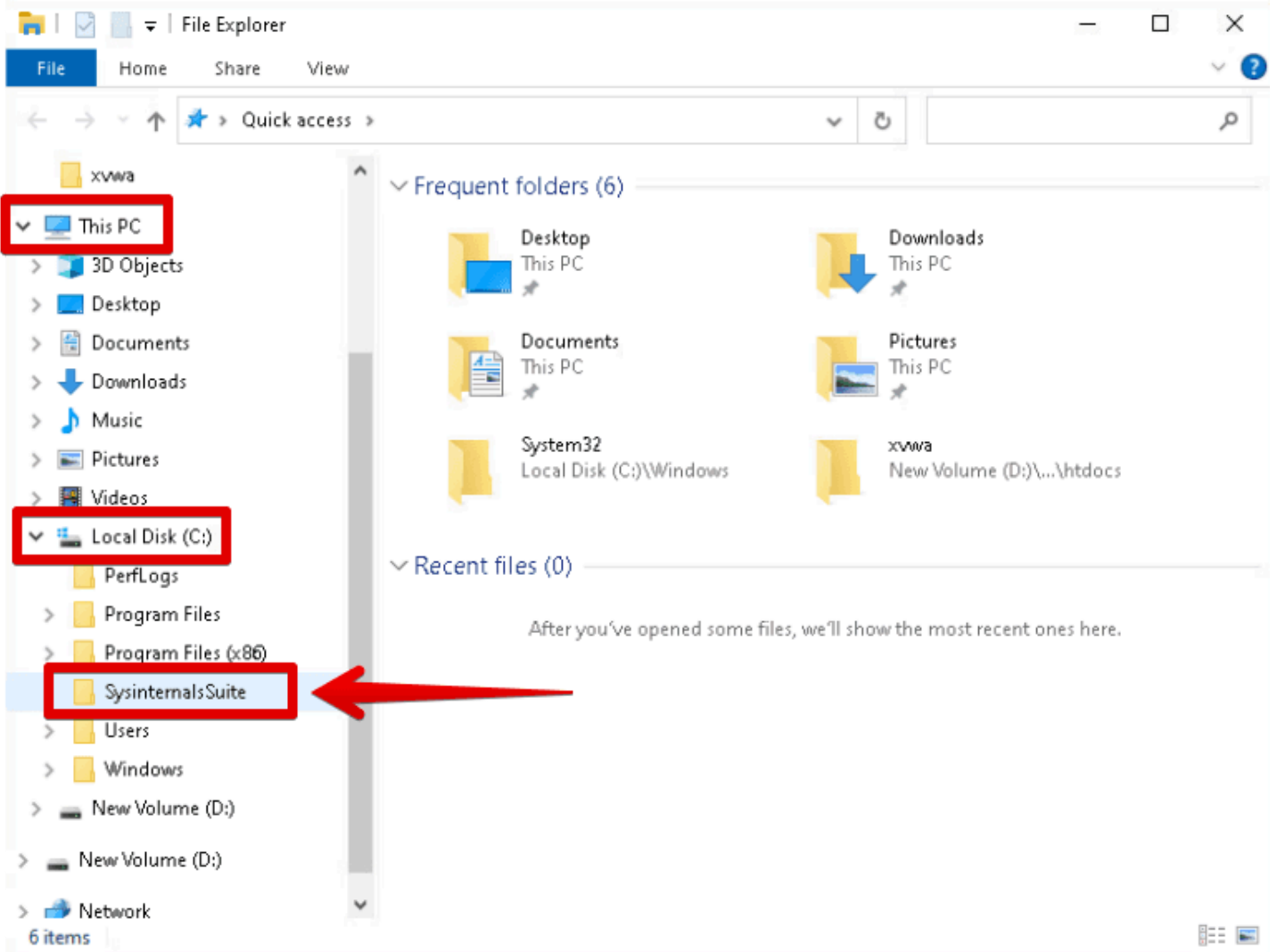
Close the **Task Manager window**.



- 14 Open File Explorer from the Taskbar.

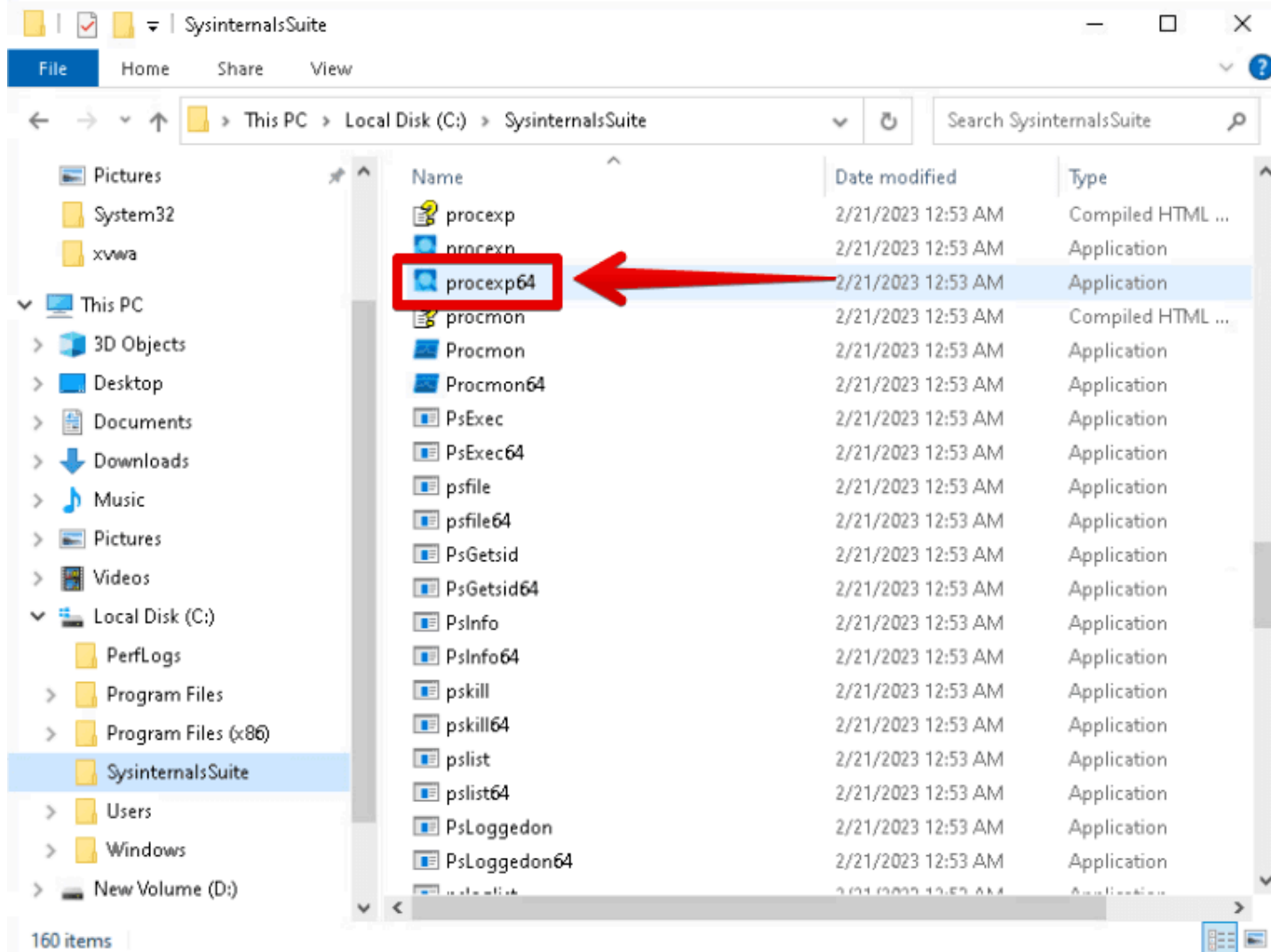


- 15 In File Explorer, **expand** This PC > Local Disk (C:) and **select** the SysinternalsSuite folder.

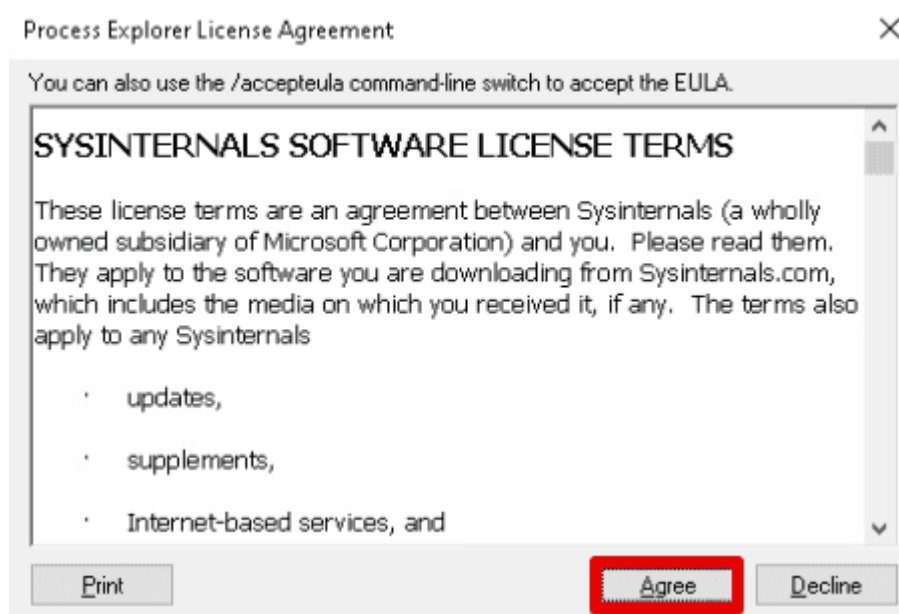


16

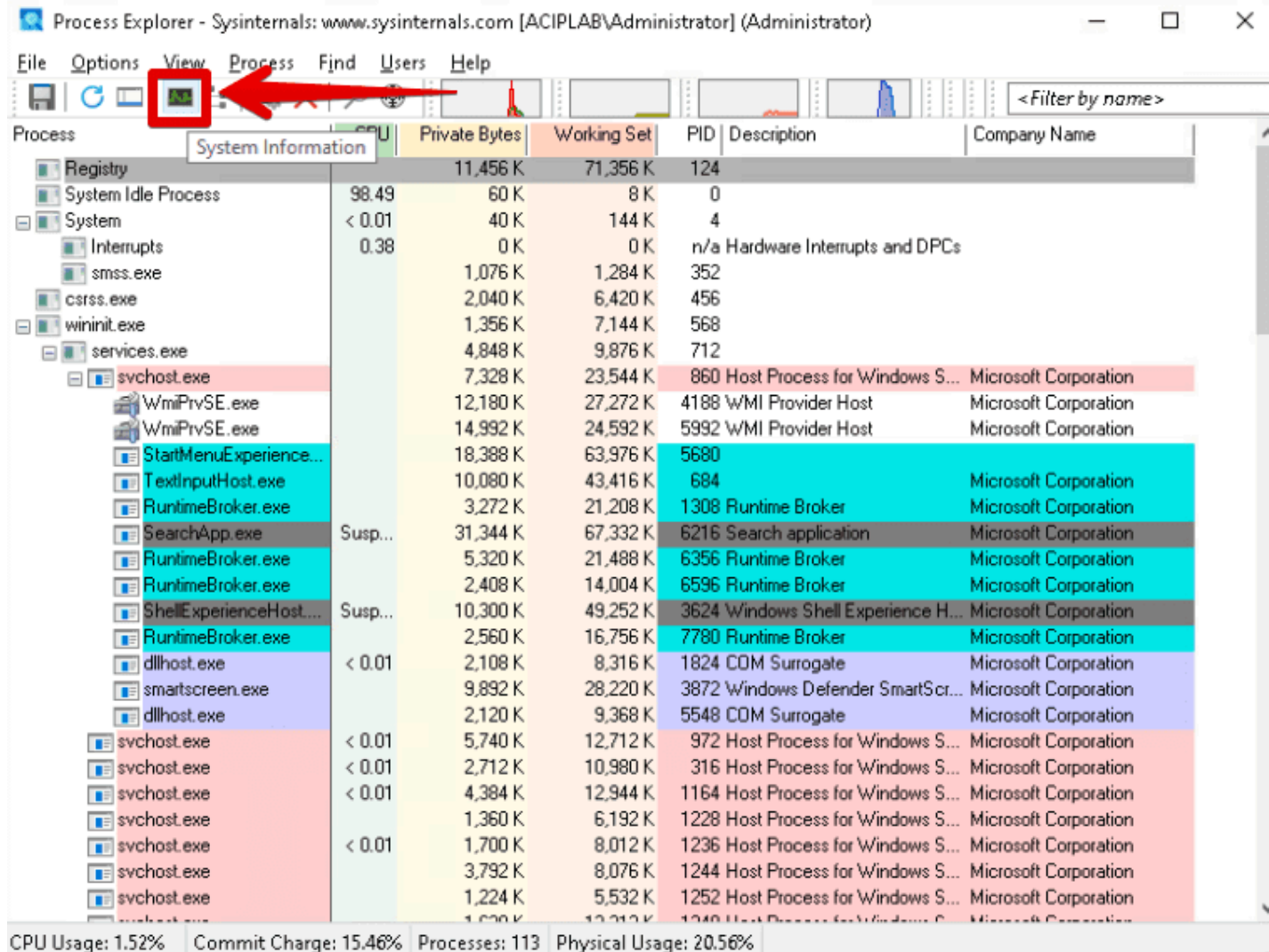
In File Explorer in the SysinternalsSuite folder, **scroll down** and **double-click** the procexp64 application.



- 17 In the **Process Explorer License Agreement** pop-up window, **click** **Agree**.



- 18 In the **Process Explorer** window, **select** **System Information** from the **menu bar**.



Note: The **Process Explorer** opens by default with all the processes running on the device. A Cyber Security Specialist can monitor these processes to ensure no malicious processes are running. The application shows a lot more information than the native **Task Manager** application.

19

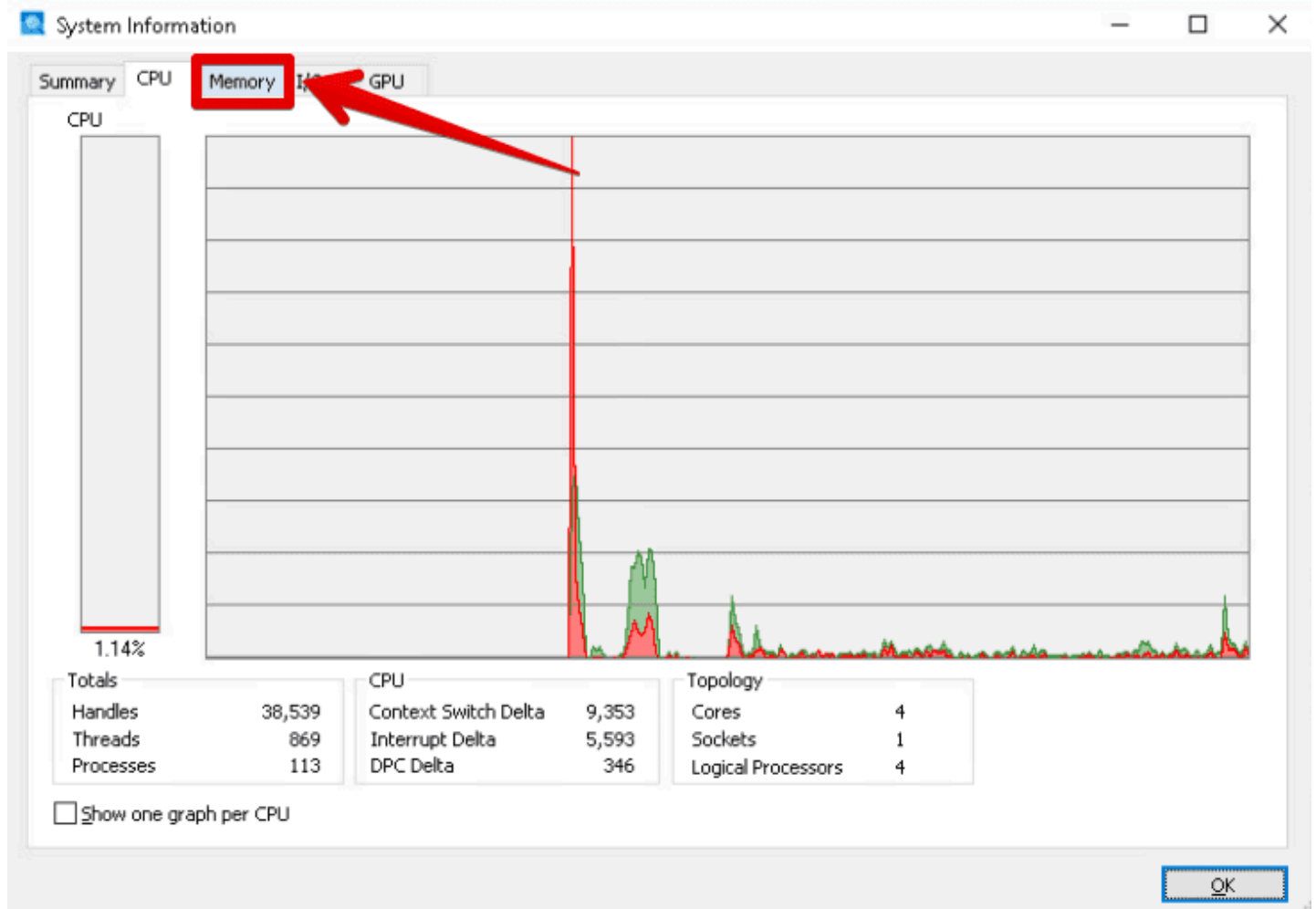
Click the **CPU** tab on the **System Information** window.



Note: The **System Information window** displays the resource utilization of the device. More detailed information can be seen on the different tabs.

20

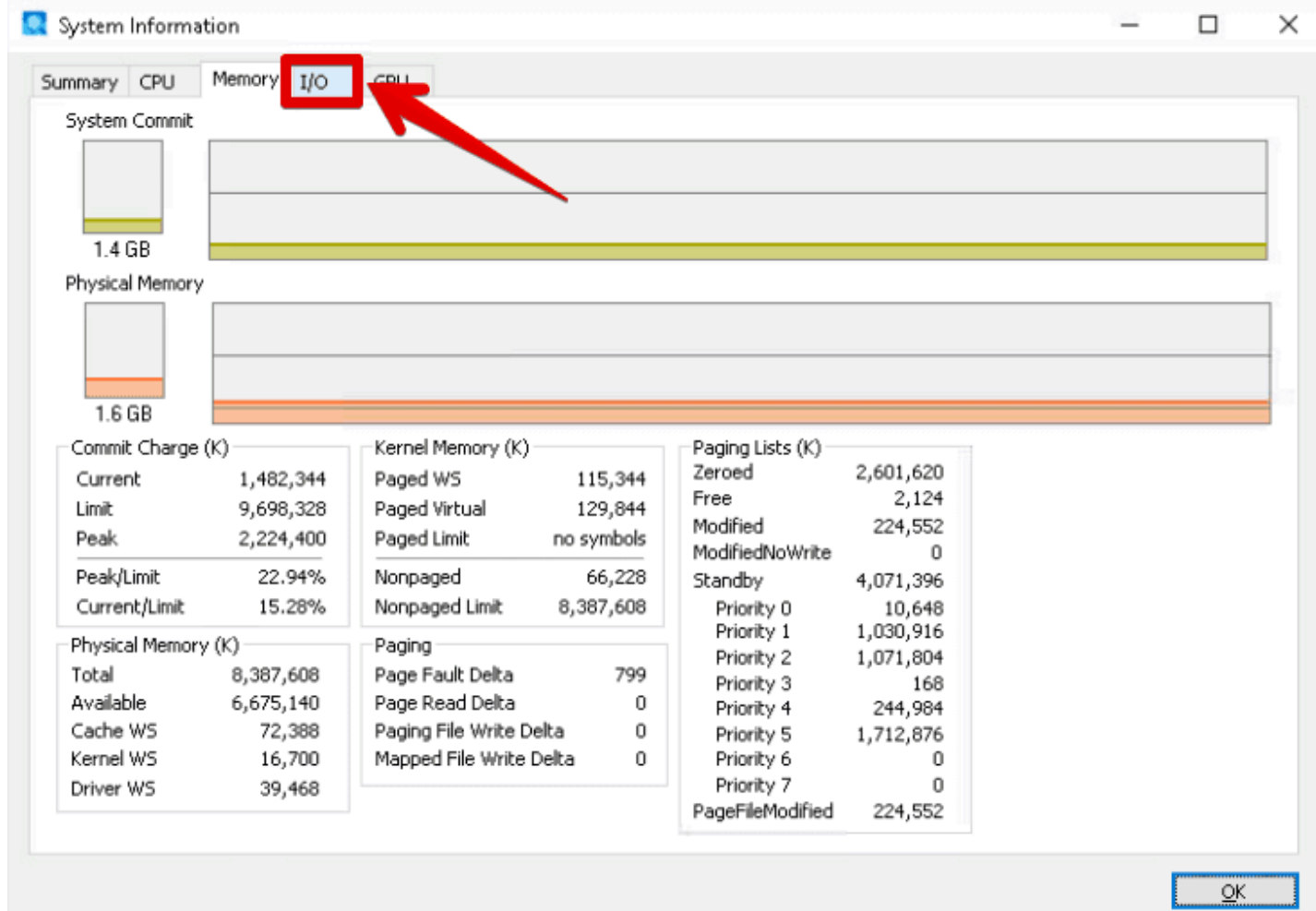
Click the **Memory tab** on the **System Information window**.



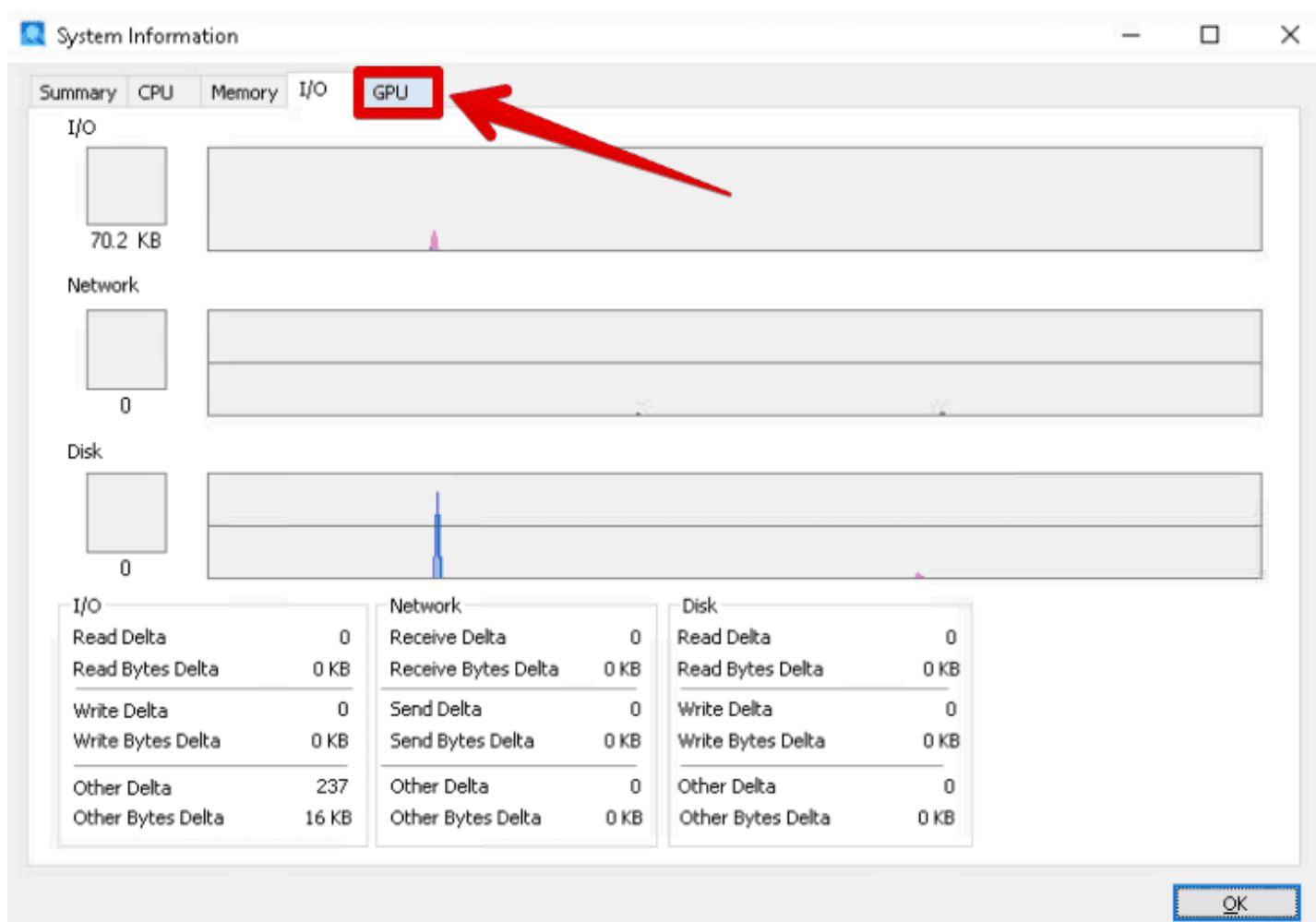
Note: A more detailed result is shown of the memory utilization of the device.

21

Select the **I/O** tab in the **System Information** window.

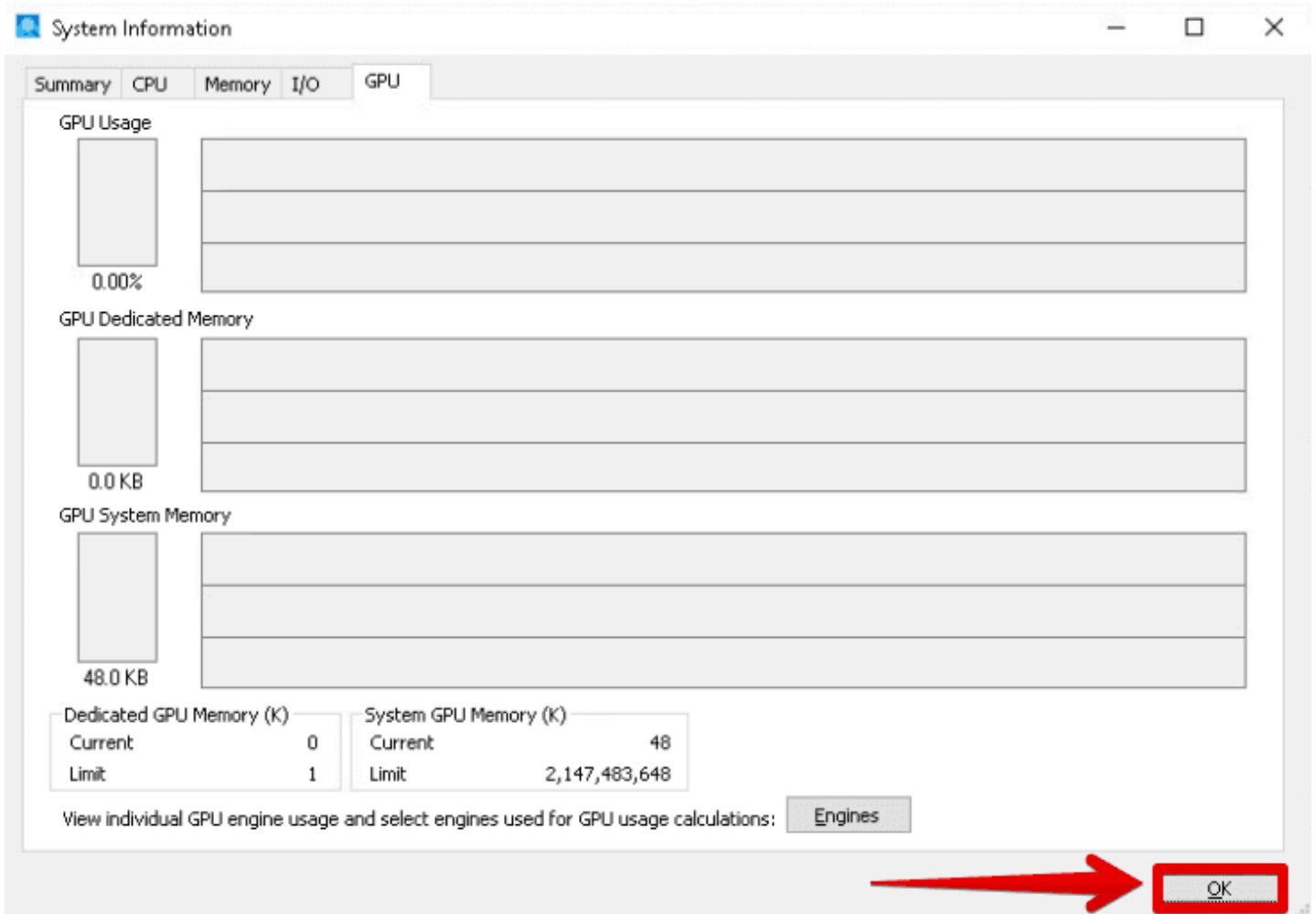


22 Select the GPU tab in the System Information window.



23

Click **OK** in the **System Information** window.



Note: No information on the **GPU tab** is available as this is a virtual device with no GPUs allocated. A Cyber Security Analyst will analyze GPU utilization to determine if any malicious services are running, for example, a Bitcoin miner.

24

Select the **SearchApp.exe** process in the **Process Explorer** window.

Process Explorer - Sysinternals: www.sysinternals.com [ACIPLAB\Administrator] (Administrator)

File Options View Process Find Users Help

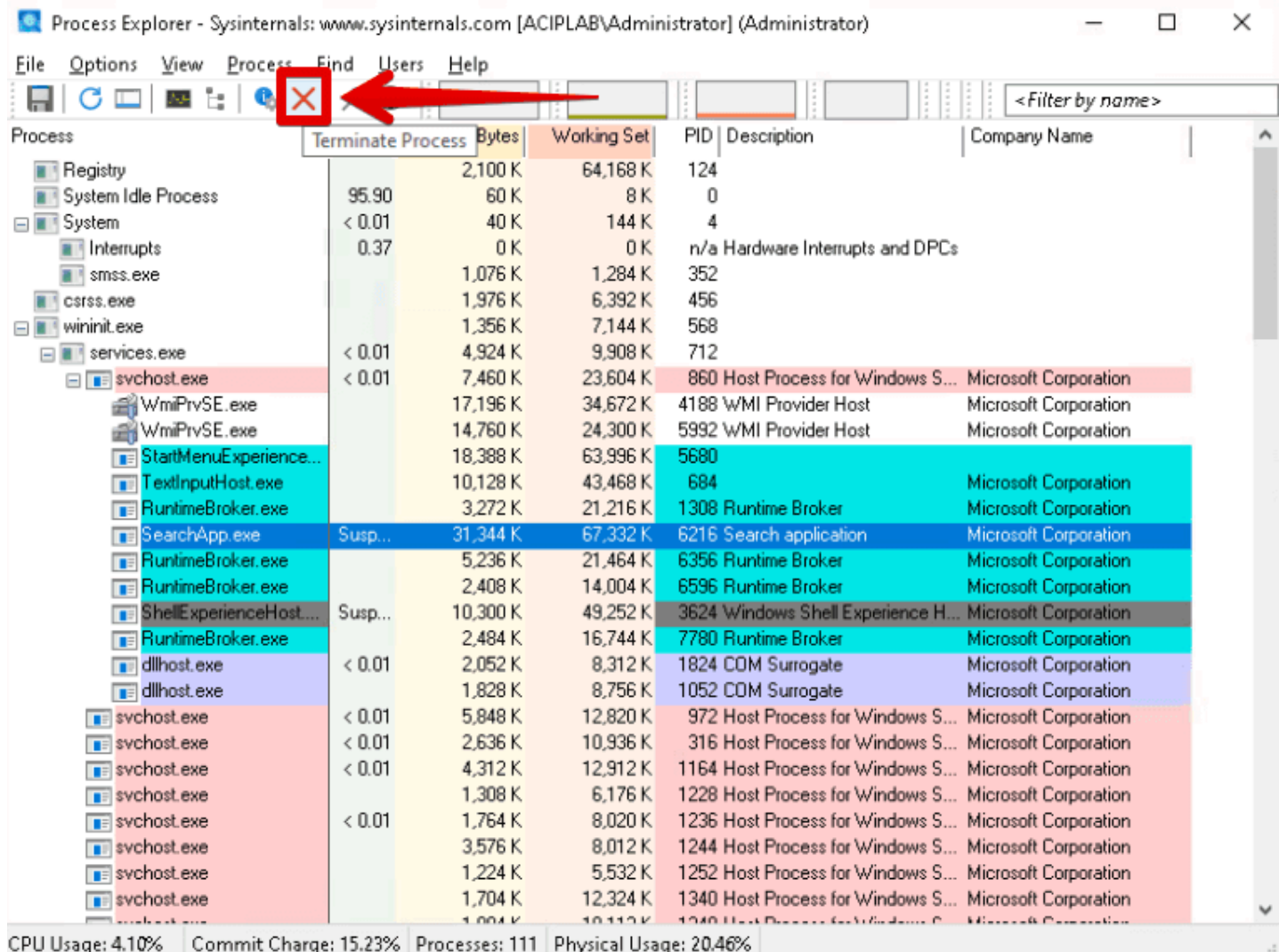
< Filter by name >

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Registry		2,100 K	64,168 K	124		
System Idle Process	100.00	60 K	8 K	0		
System	< 0.01	40 K	144 K	4		
Interrupts	< 0.01	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,076 K	1,284 K	352		
csrss.exe		1,976 K	6,404 K	456		
wininit.exe		1,356 K	7,144 K	568		
services.exe		4,924 K	9,908 K	712		
svchost.exe		7,408 K	23,592 K	860	Host Process for Windows S...	Microsoft Corporation
WmiPrvSE.exe		17,004 K	34,448 K	4188	WMI Provider Host	Microsoft Corporation
WmiPrvSE.exe		14,760 K	24,300 K	5992	WMI Provider Host	Microsoft Corporation
StartMenuExperience...		18,484 K	64,052 K	5680		
TextInputHost.exe		10,128 K	43,460 K	684		Microsoft Corporation
RuntimeBroker.exe		3,348 K	21,232 K	1308	Runtime Broker	Microsoft Corporation
SearchApp.exe	Susp...	31,344 K	67,332 K	6216	Search application	Microsoft Corporation
RuntimeBroker.exe		5,304 K	21,480 K	6396	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		2,408 K	14,004 K	6596	Runtime Broker	Microsoft Corporation
ShellExperienceHost...	Susp...	10,300 K	49,252 K	3624	Windows Shell Experience H...	Microsoft Corporation
RuntimeBroker.exe		2,484 K	16,744 K	7780	Runtime Broker	Microsoft Corporation
dllhost.exe		2,052 K	8,312 K	1824	COM Surrogate	Microsoft Corporation
WmiPrvSE.exe		2,476 K	9,920 K	5812	WMI Provider Host	Microsoft Corporation
WmiPrvSE.exe		5,272 K	13,128 K	1484	WMI Provider Host	Microsoft Corporation
dllhost.exe		2,240 K	9,344 K	1272	COM Surrogate	Microsoft Corporation
svchost.exe	< 0.01	5,840 K	12,816 K	972	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	2,584 K	10,924 K	316	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	4,312 K	12,924 K	1164	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,308 K	6,176 K	1228	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	1,764 K	8,016 K	1236	Host Process for Windows S...	Microsoft Corporation
svchost.exe		3,576 K	8,012 K	1244	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,324 K	5,520 K	1252	Host Process for Windows S...	Microsoft Corporation

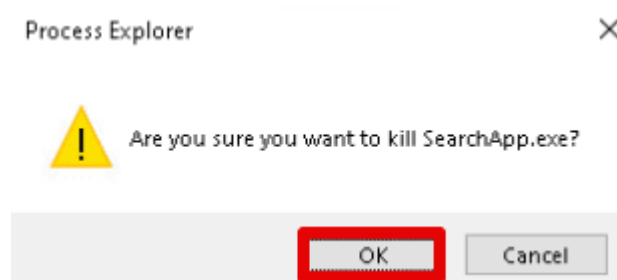
CPU Usage: 0.76% Commit Charge: 15.37% Processes: 114 Physical Usage: 20.60%

25

Select **Terminate Process** in the menu bar in the Process Explorer window.



26 Click OK on the Process Explorer pop-up window.



Note: If a malicious process is detected, a Cyber Security Specialist can terminate the specific process and then remove the malicious software.

27 Close the Process Explorer and File Explorer windows.

Process Explorer - Sysinternals: www.sysinternals.com [ACIPLAB\Administrator] (Administrator)

File Options View Process Find Users Help

< Filter by name >

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Registry		2,068 K	64,332 K	124		
System Idle Process	97.27	60 K	8 K	0		
System	< 0.01	40 K	144 K	4		
Interrupts	0.38	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,076 K	1,284 K	352		
csrss.exe		1,976 K	6,396 K	456		
wininit.exe		1,356 K	7,144 K	568		
services.exe		4,880 K	9,904 K	712		
svchost.exe		7,756 K	23,832 K	860	Host Process for Windows S...	Microsoft Corporation
WmiPrvSE.exe		17,612 K	34,684 K	4188	WMI Provider Host	Microsoft Corporation
WmiPrvSE.exe		14,508 K	24,244 K	5992	WMI Provider Host	Microsoft Corporation
StartMenuExperience...		18,448 K	64,056 K	5680		
TextInputHost.exe		10,128 K	43,484 K	684		Microsoft Corporation
RuntimeBroker.exe		3,272 K	21,216 K	1308	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe	< 0.01	5,892 K	23,216 K	6356	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		2,544 K	14,040 K	6596	Runtime Broker	Microsoft Corporation
ShellExperienceHost...	Susp...	10,300 K	49,252 K	3624	Windows Shell Experience H...	Microsoft Corporation
RuntimeBroker.exe		2,484 K	16,744 K	7780	Runtime Broker	Microsoft Corporation
dllhost.exe		2,052 K	8,312 K	1824	COM Surrogate	Microsoft Corporation
dllhost.exe	< 0.01	2,176 K	9,276 K	7300	COM Surrogate	Microsoft Corporation
SearchApp.exe	Susp...	25,856 K	65,912 K	4424	Search application	Microsoft Corporation
svchost.exe	< 0.01	5,776 K	12,800 K	972	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	2,584 K	10,924 K	316	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	4,296 K	12,904 K	1164	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,312 K	6,180 K	1228	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	1,712 K	8,004 K	1236	Host Process for Windows S...	Microsoft Corporation
svchost.exe		3,576 K	8,012 K	1244	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,224 K	5,532 K	1252	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,848 K	12,384 K	1340	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,004 K	10,112 K	1340	Host Process for Windows S...	Microsoft Corporation

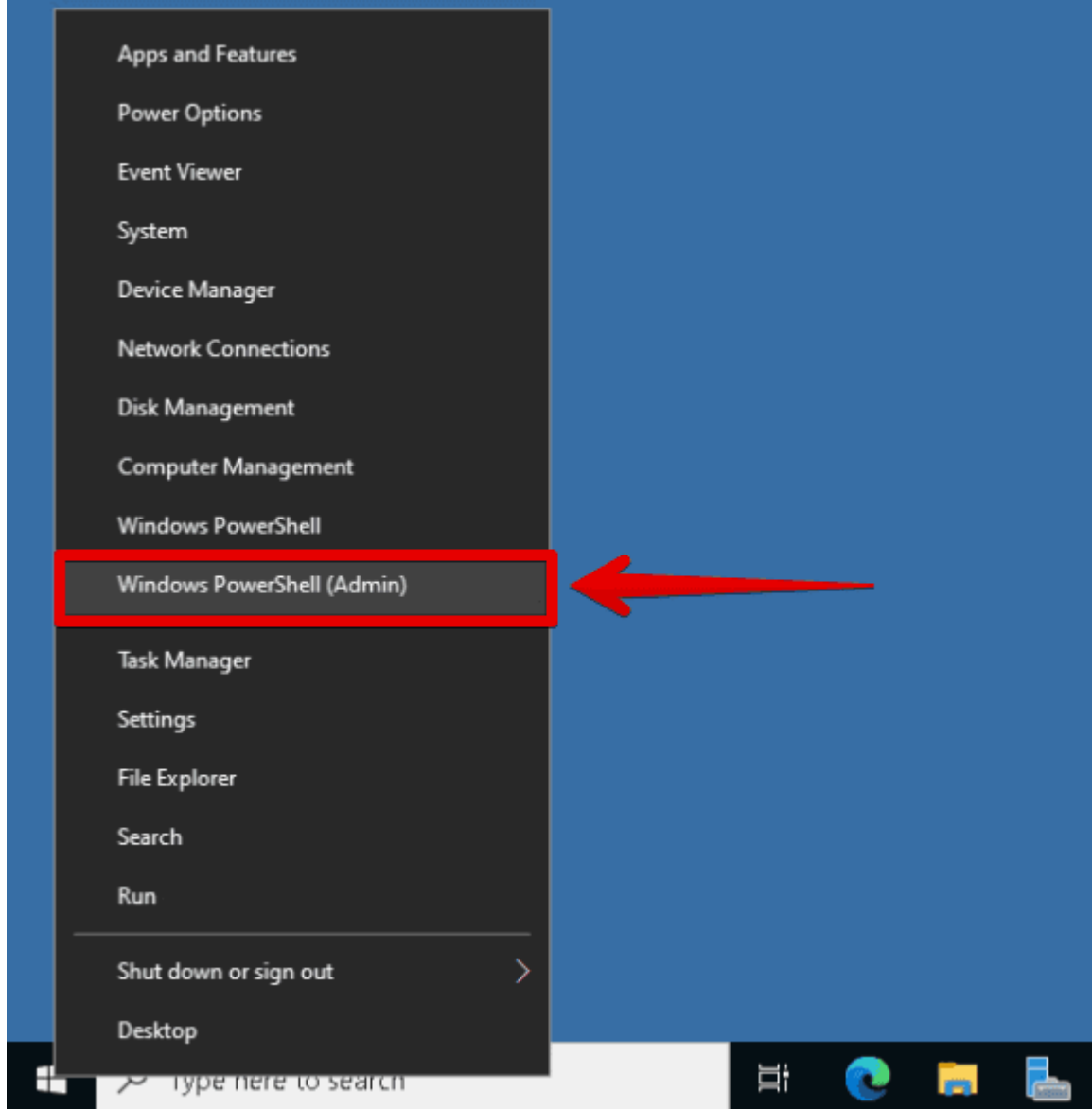
CPU Usage: 2.30% Commit Charge: 15.20% Processes: 111 Physical Usage: 20.37%

Task 2 – Monitor Resource Utilization on a Linux Device

As a Cyber Security Specialist, it is essential to monitor the resource utilization of devices on the network.

In this task, the resource utilization on a Linux device will be monitored.

- 1 **Connect** to **ACIDM01**, **right-click Start** and **select Windows PowerShell (Admin)**.



2

In the **Administrator: Windows PowerShell** window, **type** the following:

```
ssh aciadmin@192.168.0.4
```

Press Enter.

Type **yes** and **press Enter.**

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.

PS C:\Users\Administrator\ACIPLAB> ssh aciadmin@192.168.0.4
The authenticity of host '192.168.0.4 (192.168.0.4)' can't be established.
ECDSA key fingerprint is SHA256:IyfbGKHbg6z4SwEge1w/QRfawbvleRYad8dAtdgyVHQ.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.4' (ECDSA) to the list of known hosts.
aciadmin@192.168.0.4's password:
```

3

Enter the following and press Enter:

Passw0rd

```
aciadmin@localhost:~  
Windows PowerShell  
Copyright (C) Microsoft Corporation. All rights reserved.  
  
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows  
  
Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.  
  
PS C:\Users\Administrator\ACIPLAB> ssh aciadmin@192.168.0.4  
The authenticity of host '192.168.0.4 (192.168.0.4)' can't be established.  
ECDSA key fingerprint is SHA256:IyfbGKHbg6z4SwEge1w/QRfawbvleRYad8dAtdgyVHQ.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.0.4' (ECDSA) to the list of known hosts.  
aciadmin@192.168.0.4's password: Password  
Activate the web console with: systemctl enable --now cockpit.socket  
  
Last login: Sat Apr 27 00:34:03 2024  
[aciadmin@localhost ~]$
```

4

Type the following command in the aciadmin@local:~ window and press Enter:

top -n 10

```
[aciadmin@localhost ~]$ top -n 10
```

```

top - 01:24:05 up 50 min, 2 users, load average: 0.02, 0.02, 0.00
Tasks: 245 total, 1 running, 244 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.0 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 1770.3 total, 330.1 free, 959.7 used, 649.0 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 810.6 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1301	apache	20	0	1669116	13056	5816	S	0.3	0.7	0:00.43	httpd
2573	aciadmin	20	0	226040	4300	3416	R	0.3	0.2	0:00.05	top
1	root	20	0	171984	15988	10368	S	0.0	0.9	0:00.97	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_par_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	netns
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
9	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:1H-events_highpri
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	mm_percpu_wq
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthre
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace
15	root	20	0	0	0	0	S	0.0	0.0	0:00.00	ksoftirqd/0
16	root	20	0	0	0	0	I	0.0	0.0	0:00.07	rcu_preempt
17	root	rt	0	0	0	0	S	0.0	0.0	0:00.00	migration/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
21	root	rt	0	0	0	0	S	0.0	0.0	0:00.12	migration/1
22	root	20	0	0	0	0	S	0.0	0.0	0:00.01	ksoftirqd/1
24	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-events_highpri
26	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
27	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	inet_frag_wq
28	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kauditd
29	root	20	0	0	0	0	S	0.0	0.0	0:00.01	khungtaskd
30	root	20	0	0	0	0	S	0.0	0.0	0:00.00	oom_reaper
31	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	writeback
32	root	20	0	0	0	0	S	0.0	0.0	0:00.03	kcompactd0
33	root	25	5	0	0	0	S	0.0	0.0	0:00.00	ksmd
34	root	39	19	0	0	0	S	0.0	0.0	0:00.18	khugepaged
35	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	cryptd
36	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kintegrityd
37	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kblockd
38	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	blkcg_punt_bio
39	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	tpm_dev_wq
40	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	md



Note: Executing the command displays the first 10 number of repetition processes.

5

Press 'q' to exit the top application.

```
aciadmin@localhost:~
Tasks: 245 total, 1 running, 244 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.3 us, 0.0 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 1770.3 total, 330.1 free, 959.7 used, 649.0 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 810.6 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 1695 aciadmin  20   0 3900564 276232 120704 S   0.7  15.2   0:03.06 gnome-shell
 2573 aciadmin  20   0 226040  4300   3416 R   0.3   0.2   0:00.07 top
    1 root      20   0 171984  15988 10368 S   0.0   0.9   0:00.97 systemd
    2 root      20   0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 rcu_gp
    4 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
    5 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 netns
    7 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
    9 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 kworker/0:1H-events_highpri
   10 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
   12 root      20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthre
   13 root      20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude
   14 root      20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace
   15 root      20   0      0      0      0 S   0.0   0.0   0:00.00 ksoftirqd/0
   16 root      20   0      0      0      0 I   0.0   0.0   0:00.07 rcu_preempt
   17 root      rt   0      0      0      0 S   0.0   0.0   0:00.00 migration/0
   19 root      20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
   20 root      20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
   21 root      rt   0      0      0      0 S   0.0   0.0   0:00.12 migration/1
   22 root      20   0      0      0      0 S   0.0   0.0   0:00.01 ksoftirqd/1
   24 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
   26 root      20   0      0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
   27 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 inet_frag_wq
   28 root      20   0      0      0      0 S   0.0   0.0   0:00.00 kauditd
   29 root      20   0      0      0      0 S   0.0   0.0   0:00.01 khungtaskd
   30 root      20   0      0      0      0 S   0.0   0.0   0:00.00 oom_reaper
   31 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 writeback
   32 root      20   0      0      0      0 S   0.0   0.0   0:00.03 kcompactd0
   33 root      25   5      0      0      0 S   0.0   0.0   0:00.00 ksmd
   34 root      39  19      0      0      0 S   0.0   0.0   0:00.18 khugepaged
   35 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 cryptd
   36 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 kintegrityd
   37 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 kblockd
   38 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 blkcg_punt_bio
   39 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 tpm_dev_wq
   40 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 md
   41 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 edac-poller
   42 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 watchdogd
   44 root      20   0      0      0      0 S   0.0   0.0   0:00.00 kswapd0
   49 root      0 -20     0      0      0 I   0.0   0.0   0:00.00 kthrotld
   56 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 irq/24-pciehp
   57 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 irq/25-pciehp
   58 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 irq/26-pciehp
   59 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 irq/27-pciehp
   60 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 irq/28-pciehp
   61 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 irq/29-pciehp
   62 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 irq/30-pciehp
   63 root     -51   0      0      0      0 S   0.0   0.0   0:00.00 irq/31-pciehp
[aciadmin@localhost ~]$
```

Type `clear` and press `Enter` to clear the screen.

6

Enter the following command in the `aciadmin@local:~` window and press `Enter`:

```
top -u aciadmin
```

```
[aciadmin@localhost ~]$ top -u aciadmin
```



Note: Executing the command displays the processes for a specific user. This can be useful to determine if malicious applications are running on the system. From the

output, several types of information are available that can be used to investigate resource utilization.

7

Press 'q' to exit the top application.

```
aciadmin@localhost:~  
Tasks: 243 total, 2 running, 241 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 0.2 us, 0.2 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st  
MiB Mem : 1770.3 total, 330.1 free, 959.7 used, 649.0 buff/cache  
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 810.6 avail Mem  
  
  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM    TIME+  COMMAND  
 1695 aciadmin  20   0 3900564 276168 120704 S   0.3   15.2   0:03.11 gnome-shell  
 1552 aciadmin  20   0   22476   14108   10336 S   0.0    0.8   0:00.29 systemd  
 1554 aciadmin  20   0   25836   5588     4 S   0.0    0.3   0:00.00 (sd-pam)  
 1566 aciadmin  20   0 452396   8056   5084 S   0.0    0.4   0:00.02 gnome-keyring-d  
 1570 aciadmin  20   0 374144   5792   5296 S   0.0    0.3   0:00.00 gdm-wayland-ses  
 1572 aciadmin  20   0   10664   4652   4180 S   0.0    0.3   0:00.01 dbus-broker-lau  
 1575 aciadmin  20   0    6356   4096   2428 S   0.0    0.2   0:00.11 dbus-broker  
 1578 aciadmin  20   0 513096  17236  15208 S   0.0    1.0   0:00.03 gnome-session-b  
 1667 aciadmin  20   0 303788   4976   4536 S   0.0    0.3   0:00.00 gnome-session-c  
 1668 aciadmin  20   0 808552  18516  15748 S   0.0    1.0   0:00.08 gnome-session-b  
 1856 aciadmin  20   0 452156   7784   6916 S   0.0    0.4   0:00.01 gvfsd  
 1861 aciadmin  20   0 379932   8236   5564 S   0.0    0.5   0:00.00 gvfsd-fuse  
 1868 aciadmin  20   0 308584   9520   6776 S   0.0    0.5   0:00.00 at-spi-bus-laun  
 1873 aciadmin  20   0   10532   4368   4012 S   0.0    0.2   0:00.00 dbus-broker-lau  
 1874 aciadmin  20   0    4980   3004   2700 S   0.0    0.2   0:00.00 dbus-broker  
 1901 aciadmin  20   0 448100   7164   4696 S   0.0    0.4   0:00.00 xdg-permission-  
 1903 aciadmin  20   0 585252  21056  18184 S   0.0    1.2   0:00.03 gnome-shell-cal  
 1911 aciadmin  20   0 606140  26740  22564 S   0.0    1.5   0:00.05 evolution-sourc  
 1915 aciadmin   9 -11 257808  10560   7348 S   0.0    0.6   0:00.02 pipewire  
 1916 aciadmin   9 -11 543880  14924  11936 S   0.0    0.8   0:00.04 wireplumber  
 1917 aciadmin   9 -11 237120   7556   6124 S   0.0    0.4   0:00.00 pipewire-pulse  
 1921 aciadmin  20   0 895780  37932  29244 S   0.0    2.1   0:00.04 goa-daemon  
 1934 aciadmin  20   0 1050064 30348  24220 S   0.0    1.7   0:00.07 evolution-calen  
 1951 aciadmin  20   0 527532  10840   7532 S   0.0    0.6   0:00.00 gvfs-udisks2-vo  
 1970 aciadmin  20   0 530996  11460   8228 S   0.0    0.6   0:00.19 goa-identity-se  
 1973 aciadmin  20   0 448188   5492   4996 S   0.0    0.3   0:00.00 gvfs-mtp-volume  
 1985 aciadmin  20   0 155860   4828   4428 S   0.0    0.3   0:00.00 dconf-service  
 1986 aciadmin  20   0 879672  27672  22256 S   0.0    1.5   0:00.04 evolution-addre  
 1989 aciadmin  20   0 450016   6136   5524 S   0.0    0.3   0:00.00 gvfs-gphoto2-vo  
 1995 aciadmin  20   0 448284   8400   5840 S   0.0    0.5   0:00.00 gvfs-goa-volume  
 2022 aciadmin  20   0 161724   9296   6584 S   0.0    0.5   0:00.01 at-spi2-registr  
 2023 aciadmin  20   0 2722692 26492  21304 S   0.0    1.5   0:00.04 gjs  
 2034 aciadmin  20   0 522124   6332   5752 S   0.0    0.3   0:00.00 gsd-a11y-settin  
 2035 aciadmin  20   0 666848  29404  23472 S   0.0    1.6   0:00.10 gsd-color  
 2036 aciadmin  20   0 594320  20556  16020 S   0.0    1.1   0:00.02 gsd-datetime  
 2044 aciadmin  20   0 524492   9504   6776 S   0.0    0.5   0:00.04 gsd-housekeepin  
 2045 aciadmin  20   0 665168  28312  22564 S   0.0    1.6   0:00.08 gsd-keyboard  
 2047 aciadmin  20   0 735956  33224  24920 S   0.0    1.8   0:00.10 gsd-media-keys  
 2052 aciadmin  20   0 666164  29304  23364 S   0.0    1.6   0:00.10 gsd-power  
 2053 aciadmin  20   0 466836  16992  11164 S   0.0    0.9   0:00.01 gsd-print-notif  
 2057 aciadmin  20   0 669616   6244   5608 S   0.0    0.3   0:00.01 gsd-rfkill  
 2058 aciadmin  20   0 448040   6020   5480 S   0.0    0.3   0:00.00 gsd-screensaver  
 2064 aciadmin  20   0 677680   9484   8404 S   0.0    0.5   0:00.01 gsd-sharing  
 2072 aciadmin  20   0 602912  11468  10088 S   0.0    0.6   0:00.16 gsd-smartcard  
 2073 aciadmin  20   0 530664  10992   7960 S   0.0    0.6   0:00.01 gsd-sound  
 2090 aciadmin  20   0 671460   9120   6400 S   0.0    0.5   0:00.00 gsd-usb-protect  
 2092 aciadmin  20   0 665836  28648  22536 S   0.0    1.6   0:00.10 gsd-wacom  
 2124 aciadmin  20   0 232036   6012   5316 S   0.0    0.3   0:00.00 gsd-disk-utilit  
[aciadmin@localhost ~]$
```

Type **clear** and press **Enter** to clear the screen.

8

Type the following and press **Enter** in the **aciadmin@local:~** window:

9

In the `aciadmin@local:~` window, with the `top` application running, **press** the 'z' key.



Note: Pressing the 'z' key changes the color of the output, making the results more visible.

10

Press 'q' to exit the top application.

Type `clear` and **press** `Enter` to clear the screen.

11

Type the following in the `aciadmin@local:~` window:

```
sudo dnf install epel-release
```

Press `Enter`.

In the `password` field, **enter** the following:

```
Passw0rd
```

Press `Enter`.

Press 'y' and `Enter` when prompted.

12

Type the following and **press** `Enter` in the `aciadmin@local:~` window:

```
sudo dnf install htop -y
```

```
[aciadmin@localhost ~]$ sudo dnf install htop -y
Extra Packages for Enterprise Linux 9 - x86_64                               16 MB/s | 21 MB    00:01
Last metadata expiration check: 0:00:07 ago on Sat 27 Apr 2024 01:33:54 AM EDT.
Dependencies resolved.
=====
Package                Architecture          Version              Repository           Size
=====
Installing:
htop                   x86_64                3.3.0-1.el9          epel                  198 k
Installing dependencies:
hwloc-libs             x86_64                2.4.1-5.el9          baseos                2.1 M
=====
Transaction Summary
=====
Install 2 Packages

Total download size: 2.3 M
Installed size: 3.5 M
Downloading Packages:
(1/2): hwloc-libs-2.4.1-5.el9.x86_64.rpm                                8.6 MB/s | 2.1 MB    00:00
(2/2): htop-3.3.0-1.el9.x86_64.rpm                                      549 kB/s | 198 kB    00:00
-----
Total                                                                    3.0 MB/s | 2.3 MB    00:00
Extra Packages for Enterprise Linux 9 - x86_64                          1.0 MB/s | 1.6 kB    00:00
Importing GPG key 0x3228467C:
  Userid      : "Fedora (epel9) <epel@fedoraproject.org>"
  Fingerprint: FF8A D134 4597 106E CE81 3B91 8A38 72BF 3228 467C
  From        : /etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-9
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                                    1/1
  Installing     : hwloc-libs-2.4.1-5.el9.x86_64                      1/2
  Installing     : htop-3.3.0-1.el9.x86_64                            2/2
  Running scriptlet: htop-3.3.0-1.el9.x86_64                          2/2
  Verifying      : hwloc-libs-2.4.1-5.el9.x86_64                      1/2
  Verifying      : htop-3.3.0-1.el9.x86_64                            2/2

Installed:
  htop-3.3.0-1.el9.x86_64                hwloc-libs-2.4.1-5.el9.x86_64

Complete!
[aciadmin@localhost ~]$
```



Note: The **htop** application is another application that can be used to monitor processes on a Linux device.

Type **clear** and press **Enter** to clear the screen after the installation has been completed.

13

Type the following and press **Enter** in the **aciadmin@local:~** window:

htop

```
[aciadmin@localhost ~]$ htop.
```



Note: The processes that are currently running on the device are displayed. Notice that there are different colors for different processes.

14

Press F6 in the aciadmin@local:~ window.

15

Using the up arrow key, **navigate** to USER and **press** Enter.



Note: A filter is used to display the services that are running on the aciadmin user account.

16

Press F10 in the aciadmin@local:~ window.

17

Type exit and **press** Enter to close the ssh session.

Close the Windows PowerShell window.

Exercise 2 – Detecting Unauthorized User Privilege Escalation

A Cyber Security Analyst must be able to monitor account privileges and determine if an account has been assigned unauthorized privileges. If a user is detected that having unauthorized privileges, it might indicate that the network has been compromised.

In this exercise, auditing on user accounts will be enabled, and a user will be assigned unauthorized privileges. Event Viewer will be used to detect these unauthorized privileges.

Learning Outcomes

After completing this exercise, you should be able to:

- Enable User Privilege Monitoring
- Generate Auditing Logs
- Detect an Unauthorized User Privilege Escalation

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01** – Windows Server 2022 – Domain Controller

Task 1 – Enable User Privilege Monitoring

A Cyber Security Specialist needs to monitor the privileges that have been assigned to users on the network and remove unauthorized privileges.

In this task, user-privileged auditing will be enabled.

1

Connect to **ACIDC01**.

Click **Send Ctrl+Alt+Delete button** on the top of the screen.

Sign-in with the **following credentials**:

Username: **ACIPLAB\Administrator**

Password: **Passw0rd**

In **Server Manager**, **click** **Tools** and **select** **Group Policy Management**.

2

In the **Group Policy Management window**, **expand** **Forest: ACIPLAB.COM** > **Domains** > **ACIPLAB.COM**.

3 **Select** the **Default Domain Policy** and **click OK** on the **Group Policy Management Console** pop-up window.

4 **Right-click** the **Default Domain Policy** and **select Edit**.



Note: Editing the **Default Domain Policy** will apply to all network users and computers. In a real-world scenario, a separate Group Policy will be created and only applied to a selected group of devices or users.

5 In the **Group Policy Management Editor** window, **expand** **Computer Configuration** > **Policies** > **Windows Settings** > **Security Settings** > **Advanced Audit Policy Configuration** > **Audit Policies**.

6 **Select** **Account Management** on the left pane and **click** **Audit Security Group Management** in the right pane.

7 In the **Account Management** pane, **right-click** **Audit Security Group Management** and **select Properties**.

8 On the **Audit Security Group Management Properties** window, **enable** the **Configure** the following audit events tick box.

9

Enable the **Success and Failure** tick boxes and **click** **OK**.



Note: Enabling the **Security Management Policy** will enable the auditing of **Security groups in the Active directory**. If a user is added to the security group, a log will be created that can be viewed in Event Viewer.

10

Close the **Group Policy Management Editor** window.

11

Close the **Group Policy Management** window.

12

Right-click **Start** and **select** **Windows PowerShell (Admin)**.

13

In the **Administrator: Windows PowerShell** window, **type** the **following**:

```
gpupdate /force
```

Press **Enter**.

14

Close the **Administrator: Windows PowerShell** window.



Note: Executing the command in **Windows PowerShell** will ensure the policy that has been created is applied to the device.

Task 2 – Generating Auditing Logs

In order to determine if the applied policy is working correctly, a user account needs to be created and assigned to a privileged group.

In this task, a user will be created and added to a privileged group to generate log files.

1

Connect to ACIDC01 in Server Manager, **click** Tools and **select** Active Directory Users and Computers.

2

In the Active Directory Users and Computers window, **expand** ACIPLAB.COM and **right-click** the Marketing.

3

Select New > User in the flyout window.

4

Complete the following fields in the New Object – User pop-up window:

First name:

Last name:

User logon name:

Click Next.

5

On the New Object – User pop-up window, **complete** the following:

Password:

Confirm password:

Untick the User must change password at next logon check-box.

Click Next.

6

Click Finish on the New Object – User pop-up window.

7

Right-click the **Jeremiah ACI** user and **select** **Properties**.

8

On the **Jeremiah ACI Properties** window, **click** the **Member Of** tab.

9

On the **Member Of** tab, **select** **Add**.

10

On the **Select Groups** window, **type** the **following** in the **Enter the object names to select** pane:

Domain Admins

Click **OK**.

11

Click **OK** on the **Jeremiah ACI Properties** window.



Note: The **Jeremiah ACI user** was added to the **Domain Admins group**. In a real-world scenario, if a network account were compromised and privileges on the account were escalated, the attacker would create an obscure account, add specific privileges, and then remove the escalated privileges from the compromised account.

12

Close the **Active Directory Users and Computers** window.

Task 3 – Detecting an Unauthorized User Privilege Escalation

A Cyber Security Specialist has applied an audit policy to audit the assignment of privileges to users.

In this task, **Event Viewer** will be used to determine if users have been assigned unauthorized privileges.

- 1 **Connect** to **ACIDC01**, **right-click** **Start** and **select** **Event Viewer**.
- 2 In the **Event Viewer** window, **expand** **Windows logs** and **click** **Security**.
- 3 In the **Security** pane, **double-click** the **Event ID 4728**.
- 4 **Click** the **Details** tab on the **Event Properties - Event 4728, Microsoft Windows security auditing** pop-up window.
- 5 **Scroll down** on the **page** to view the **audit policy** results.

Click **Close** on the **Event Properties -Event 4728, Microsoft Windows security auditing** pop-up window.



Note: The **Event ID 4728** displays that the **Administrator account** has added the user **Jeremiah** to the **Domain Admins group**. This might indicate that the administrator account has been compromised and is used for privilege escalation on the network.

- 6 In the **Security** pane, **double-click** the **Event ID 4737**.

7

Click the **Details tab** on the **Event Properties -Event 4737, Microsoft Windows security auditing pop-up window**.

8

Scroll down on the **page** to view the audit policy results.

Click **Close** on the **Event Properties -Event 4737, Microsoft Windows security auditing pop-up window**.



Note: The **Event ID 4737** displays that the **Administrator account** modified the **Security group**, which might be another indication that the account has been compromised.



Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Identifying & Analyzing Malicious Activity** practice lab.

Summary

You completed the following exercises:

- Exercise 1 – Analyze Resource Utilization
- Exercise 2 – Detecting Unauthorized User Privilege Escalation

You should now be able to:

- Monitor Resource Utilization on a Windows Device
- Monitor Resource Utilization on a Linux Device
- Enable User Privilege Monitoring
- Generate Auditing Logs
- Detect an Unauthorized User Privilege Escalation

Feedback



Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.